

Seven Segment Display Interfacing with Vamon FPGA Board

Aim

To interface a seven segment display with the Vamon FPGA board by driving each segment (a to g) directly from individual board output pins, and to verify on hardware that the correct segments light up to form the required digit pattern.

GATE Question

The Boolean expression for the output f of a 4:1 multiplexer with inputs $I_0 = R$, $I_1 = R'$, $I_2 = R'$, $I_3 = R$ and select lines $S_1 = P$, $S_0 = Q$ is one of the following:

Option	Expression
A	$(P \text{ xor } Q \text{ xor } R)'$
B	$P \text{ xor } Q \text{ xor } R$
C	$P + Q + R$
D	$(P + Q + R)'$

Question Analysis

1. Select-Line to Output Mapping

$S_1 (P)$	$S_0 (Q)$	Line Selected	Output f
0	0	I_0	R
0	1	I_1	R'
1	0	I_2	R'
1	1	I_3	R

2. Sum of Products Form

$$f = P'Q'R + P'Q R' + PQ' R' + PQR$$

3. Complete Truth Table

P	Q	R	f
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0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

4. Observation and Final Expression

f is 1 only where P, Q, R contain an odd number of 1's (001, 010, 100, 111) – the defining property of the three-input XOR operation.

$f = P \text{ xor } Q \text{ xor } R \rightarrow \text{Answer: Option (B)}$

1. Components Used

Component
Vamon Board
Seven Segment Display
Jumper Wires
Linux Computer

2. Purpose of Each Component

Component	Function
Vamon Board	Generates the segment-drive signals (a-g) for the display
Seven Segment Display	Lights up the individual segments to show the digit pattern
Jumper Wires	Connects each Vamon board pin to the corresponding segment pin
Linux Computer	Runs Termux/toolchain to compile and program the board

3. Power Connections

Pin	Connection
Vamon Board 3.3V	Seven Segment Common Pins (3 & 8)

4. Vamon Board to Seven Segment Display Connections

Vamon Pin	7-Segment Pin	Segment
20	7	a
21	6	b
22	4	c
23	2	d
25	1	e
26	9	f
27	10	g

5. Execution Procedure

1. Write a small design in Termux on the Linux computer that drives pins 20, 21, 22, 23, 25, 26 and 27 to form the required segment pattern, along with a constraint file mapping these to the physical Vamon pins.
2. Synthesize the design through the SymbiFlow toolchain to produce a bitstream, then convert it to a .bin file.
3. Connect the Vamon board to the Linux computer over USB and put it into programming mode.
4. Confirm the board is detected as a serial device before flashing.
5. Flash the .bin file onto the board using the TinyFPGA programmer utility.
6. Reset the board once flashing completes so the new segment-drive logic takes effect.
7. Check the seven segment display: the segments corresponding to a logic HIGH on their driving pin should light up, and the resulting pattern should match the intended digit.

Results and Observations

After flashing and resetting the Vamon board, the seven segment display lit up according to the signals driven on pins 20, 21, 22, 23, 25, 26 and 27, with the common pins (3 & 8) held at 3.3V. The observed segment pattern on hardware matched the intended digit, confirming that the pin mapping and drive logic were correct.